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ATS A Sample initial release precondition

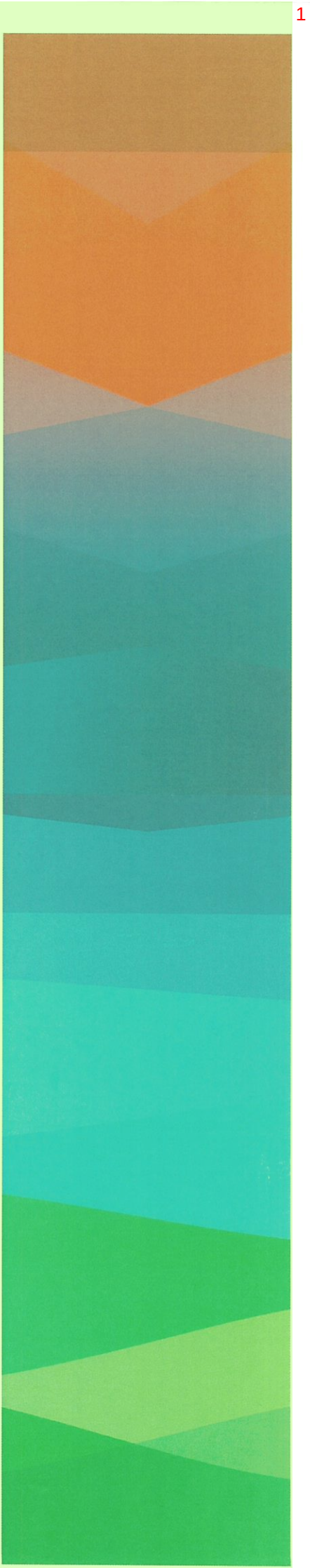
A Sample Drawing initial Release Checklist

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Pre-condition Checklist		Result/Report	Date	Resp.
Design	1. A Sample 3D model confirmed by customer?	OK	2024.9.4	卢青松
	2. A Sample Nameplate confirmed by customer? Email or drawing confirm?	OK	2024.9.4	卢青松
	3. A Sample Single part manufacturing feasibility study and drawing confirmed by supplier?	OK	2024.9.4	卢青松
	4. A sample MAT design fixed?	OK 借用R244E国产木村垫	2024.9.4	卢青松
PUR/MOE	1. A Sample Make and Buy confirmed?	OK	2024.9.4	卢青松
	2. A Sample DFMA welding process finished with MOE?	OK	2024.9.4	卢青松
	3. A Sample Canning process confirmed?	OK	2024.9.4	卢青松
Validation	1. A Sample CFD report?	OK	2024.9.4	王由

 BOSCH 博世	Design Change Review Sheet 设计变更评审表	Version 13 Effective date: 2024-09-03th
Confidential		Design Change Request No. 24_093
1. Function&Performance will be influenced? / 系统性能影响? ☐ yes/是 ☒ no/否		
If Yes, System impact evaluation? 如果是, 系统影响评估? (如果是初次样件放行, 空白处请说明样件目的) Whether need sample to test? Plan? 是否需要样件进行验证? 计划? Whether the catalyst information is confirmed? 催化剂信息是否确认? (物料号、图纸、SPEC)	初次样件放行, 用于标定测试 标定样件	System/签名 沈伟波 Catalyst/签名
2. Interface&boundary (internal and customer) will be influenced? / 接口&边界(内部和客户)影响? ☒ yes/是 ☐ no/否		
If impact customer ,whether get confirm from customer side? 请说明影响和是否得到客户的确认?	客户已校核完成	Design /签名 隋松
3. Mechanical strength&Durability will be influenced? / 机械强度&可靠性影响? ☐ yes/是 ☒ no/否		
If Yes, How to evaluate? Plan? 如果是, 请说明影响和如何评估, 计划? (例如: 仿真分析等) Whether need sample to test? Plan? 是否需要样件进行验证? 计划?	A样无需FEA仿真	Validation/签名 卢青松
4. Product document influenced?		
4.1 Interface FMEA-relevant / IFMEA:	☒ no/否 ☐ yes/是 Note	Design /签名 隋松
4.2. Product FMEA-relevant / DFMEA:	☒ no/否 ☐ yes/是 Note	
4.3. Special Charecteristics /PSC: (针对C Sample)	☒ no/否 ☐ yes/是 Note	
4.4. IMDS relevant: (针对C Sample)	☒ no/否 ☐ yes/是 Note	
4.5 Offer drawing relevant:(针对C Sample)	☒ no/否 ☐ yes/是 Note	
4.6. TCD relevant:	☒ no/否 ☐ yes/是 Note	
5. Changed parts Stock& treatment / 需要变更零件的库存统计和处理? (只针对涉及到已经生产过的零件变更, 新放行零件和本条不相关)		
☐ NA.(新零件放行,不涉及库存) ☐ 涉及到已经生产过的零件变更		
RBCW RDC(外库)和RBCW 仓库, 是否有库存,	☒ 无库存 ☐ 有库存, 库存数量	PUE /签名 李爱芳
供应商处(已做好, 未发货)是否有库存,	☒ 无库存 ☐ 有库存, 库存数量	
LPN工厂是否有库存,	☒ 无库存 ☐ 有库存, 库存数量	
在途, 运输中是否有库存	☒ 无库存 ☐ 有库存, 库存数量	
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☐ 可以继续使用完 ☐ 改制后使用 ☐ 报废	Design /签名
	☐ 改制费用; ☐ 报废成本	PUE /签名
已装车件处理方法 (发客户或标定车等) (SMP组织会议讨论)	☐ 改制; ☐ 更换; 其他(单独说明);	
6. Influence on supplier part?		
Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier? 变更后的零件, 生产可行性供应商是否确认? 反馈?		
Influence on purchasing part delivery time? 对采购件交货时间的影响?		
涉及到变更的零件, 博世已出PR, 供应商还未来得及生产的订单, 是否已经通知供应商取消. ☒ no/否 ☐ yes/是		
Does the change influence on purchasing cost ? 变更对采购成本的影响? ☐ increase/增加,金额 ☐ decrease/降低,金额 ☐ no change/不变		
PUE/签名		
7. DFMA & Sample prodction OPL?		
必要时附加DFMA 文件		
2# DFMA. 参考附件		
Design /签名 隋松		
8. Influence on LPN manufacturing&assembly process?		
Feedback from manufacturing engineer? 生产制造装配是否影响, 反馈?		
Influence on project&product delivery time? 对项目&产品交货时间的影响?		
Influence on manufacturing cost / 对生产成本的影响? ☐ increase/增加,金额 ☐ decrease/降低,金额 ☐ no change/不变		
PUE/签名		
9. PM alignment? (Influence on project Timeline? Cost?); 是否同意对项目时间和成本的影响; 需求样品订单数量;		
PM/签名		
10. 如果涉及以下情况需TCR3 审核确认。 不涉及则 NA.: ☐ ☐ 库存报废,金额; ☐ 采购成本增加,金额; ☐ 产成本增加,金额		
RBCW/TCR3		
Design Team Leader	System SM-ENG-CV1	Department Manager TCR3
Signature / 签名: 杨广松	Signature / 签名: 罗	Signature / 签名: 谢友福

2024-10-12



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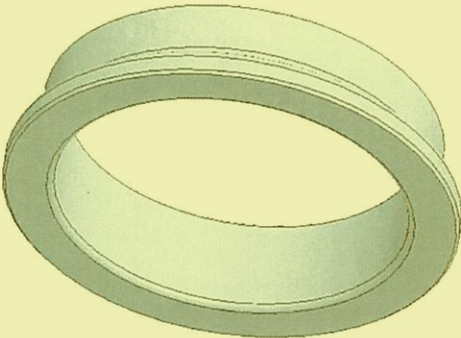
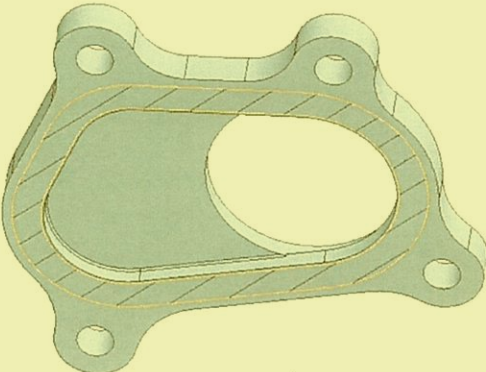
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ENG-CV

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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5102CZ-01	FLANGE	JIM-493	A	BOSCH/Customer
	变更前				

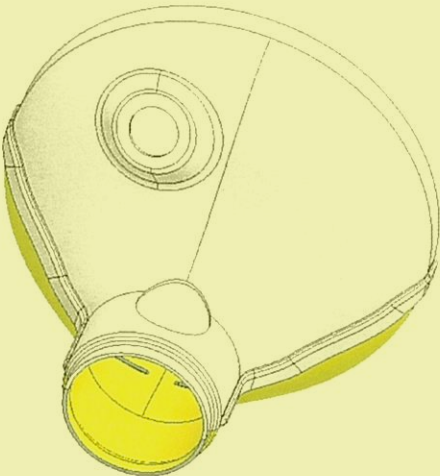
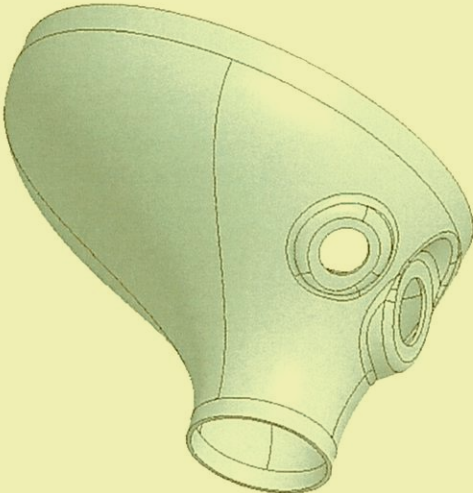
变更描述			根据客户增压器出口，重新设计连接法兰； 
	变更后		

变更原因	根据客户增压器出口，重新设计连接法兰；
风险点（影响）	法兰形状及出口大小改变，气体流速均匀性降低；
验证方法	进行CFD仿真，仿真报告如附件；

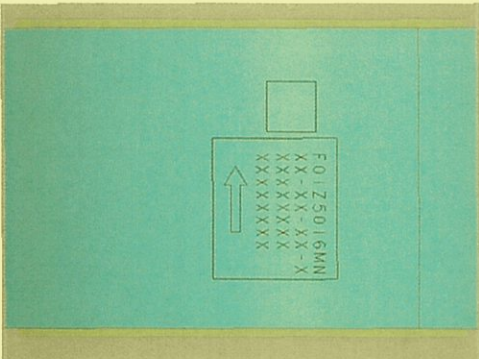
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	零件号	零件名称	Project	样件阶段	变更来源			
	F01Z5102D0-01	CONE	JIM-493	A	BOSCH/Customer			
	变更前		变更后					
变更描述								
变更原因	根据客户边界及法兰重新设计端锥;							
风险点 (影响)	端锥形状及出口大小改变, 气体流速均匀性降低;							
验证方法	进行CFD仿真, 仿真报告如附件;							

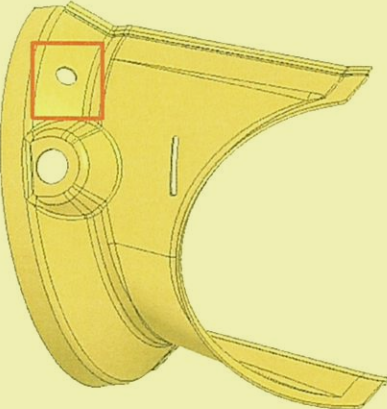
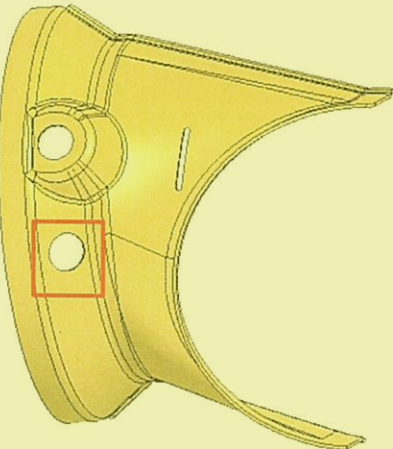
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	零件号		零件名称	Project	样件阶段	变更来源
	F01Z5017NM-00		DOC CANNING	JIM-493	A	BOSCH/Customer
			变更前	变更后		
变更描述			DOC CANNING: F01Z5016NA DOC SLEEVE: F01Z5101BG DOC CATALYST: F01Z501906 DOC MAT: F01Z510111 DOC CANNING: F01Z5017NM-00 DOC SLEEVE: F01Z5101BG DOC CATALYST: F01Z50178N-00 DOC MAT: F01Z5101PS			
变更原因	降本更换催化剂及衬垫;					
风险点 (影响)	更换衬垫催化剂有吹出风险;					
验证方法	Canning DV实验已通过;					

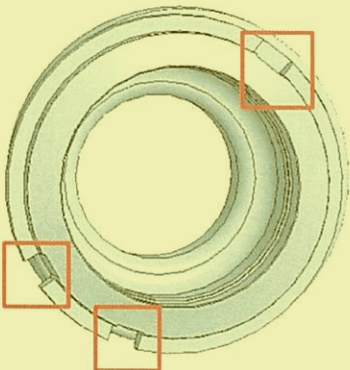
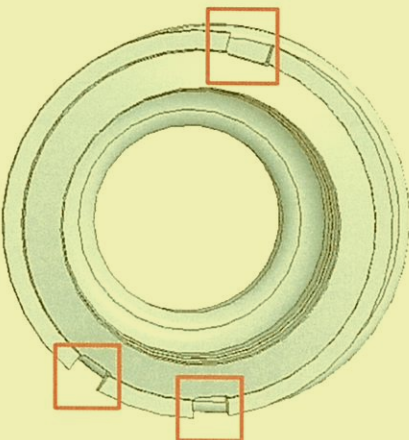
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变更描述	零件号		零件名称		Project		样件阶段		变更来源	
	F01Z510266-00		SLEEVE		JIM-493		A		BOSCH/Customer	
			变更前						变更后	
										
	更改混合器压差座位置;									
变更原因	根据客户边界重新设计压差管，供应商焊接评估在原位置无法焊接，因此更改压差传感器座的位置;									
风险点 (影响)	无									
验证方法	无									

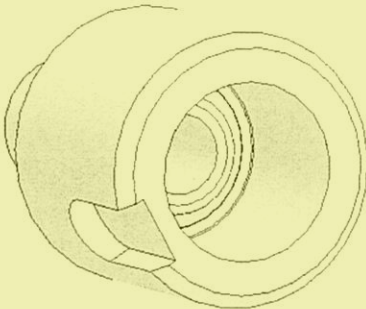
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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5102E0-00	DM flange	J1M-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
	更改尿素喷嘴座开孔角度（逆时针旋转20°）；				
变更原因	原尿素喷嘴角度会跟客户水箱干涉，尿素喷嘴更改周期长，因此修改尿素喷嘴座开槽角度（逆时针旋转20°）；				
风险点（影响）	无				
验证方法	无				

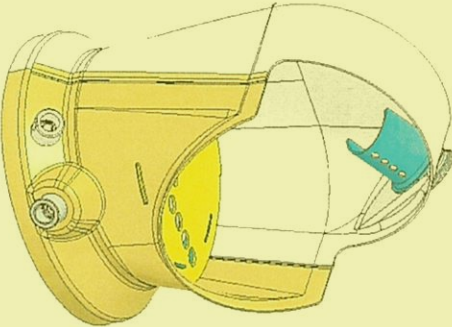
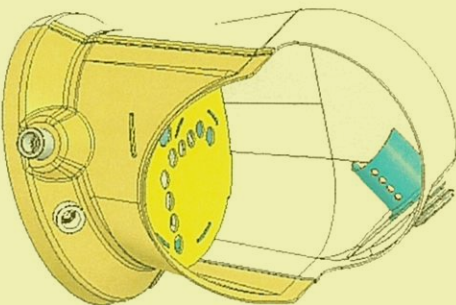
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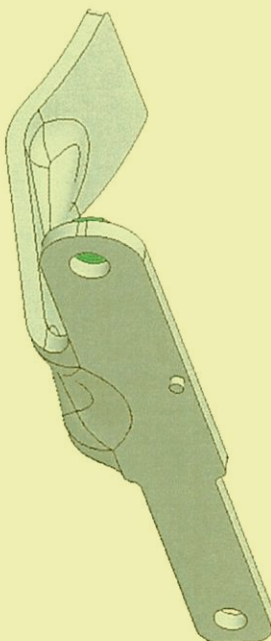
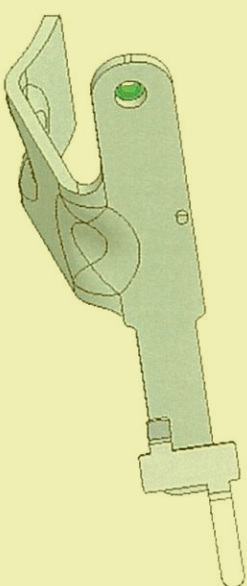
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	零件号	零件名称	Project	样件阶段	变更来源			
	F01Z510265-00	DP SENSOR BOSS	JIM-493	A	BOSCH/Customer			
	变更前							
变更描述			变更后					
根据客户边界重新设计压差管座;								
变更原因	根据客户边界重新设计压差管座;							
风险点 (影响)	无							
验证方法	无							

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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017W6-00	MIXER UNIT	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	上述混合器零件更改后，混合器组件随之更改;				
风险点（影响）	无				
验证方法	无				

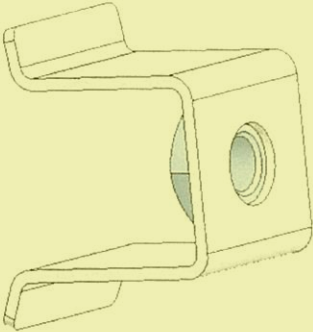
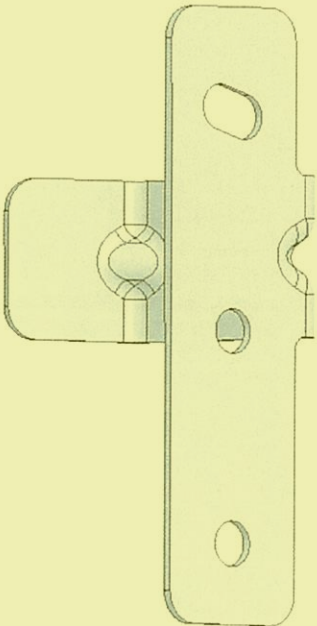
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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017X1-00	DP TUBE BRACKET WITH NUT	JIM-493	A	BOSCH/Customer
	变更前				
变更描述					
	DP TUBE BRACKE: F01Z5016MR DP TUBE BRACKET: F01Z5101BT WELDING NUT: F01Z500068		DP TUBE BRACKE: F01Z5017X1-00 DP TUBE BRACKET: F01Z5101BT DP TUBE BRACKET: F01Z5102D2-00 WELDING NUT: F01Z500068		
变更原因	客户要求新增传感器线束固定抱箍;				
风险点 (影响)	无				
验证方法	无				

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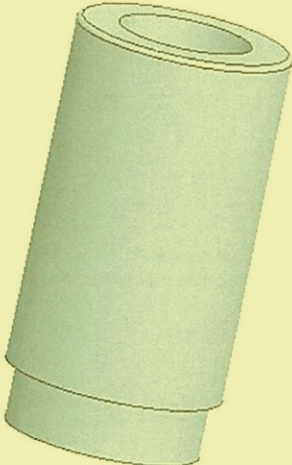
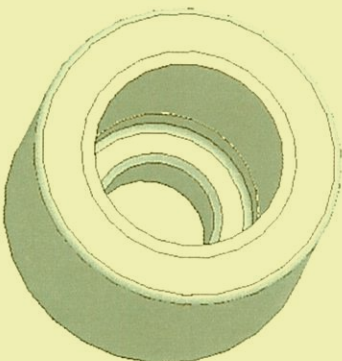
	零件号		零件名称		Project	样件阶段	变更来源
	F01Z5102DH-00		SENSOR BRACKET		JIM-493	A	BOSCH/Customer
	变更前						变更后
变更描述							
变更原因	客户要求取消水管支架，新增传感器线束固定支架；						
风险点 (影响)	无						
验证方法	无						

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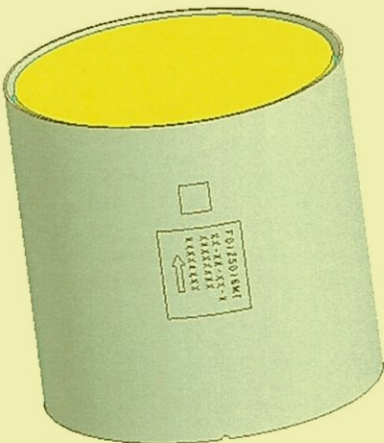
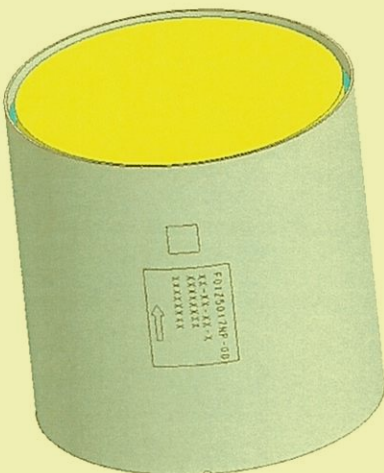
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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5102DL-00	NOx SENSOR BOSS	J1M-493	A	BOSCH/Customer
	变更前				
变更描述	变更后				
	为方便定位，NOX座新增定位台阶；				
变更原因	为方便定位，NOX座新增定位台阶；				
风险点 (影响)	无				
验证方法	无				

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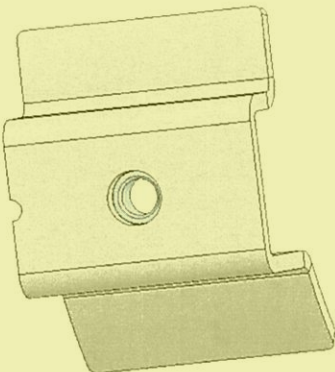
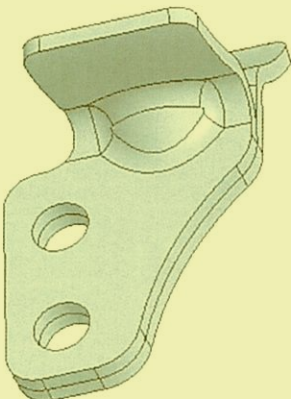
	零件号		零件名称		Project		样件阶段		变更来源	
	F01Z5101BS		TEMP-SENSOR BOSS		JIM-493		A		BOSCH/Customer	
	变更前				变更后					
变更描述										
		供应商请求，T4 T5 T6统一用一个温度传感器座；								
变更原因		统一温度传感器座，T4 T5 T6统一用一个温度传感器座；								
风险点（影响）		无								
验证方法		无								

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	零件号		零件名称	Project	样件阶段	变更来源
	F01Z5017NP-00		DPF CANNING	JIM-493	A	BOSCH/Customer
	变更前			变更后		
变更描述	 DPF CANNING: F01Z5016MT DPF SLEEVE: F01Z5101BY DPF CATALYST: F01Z501890 DPF MAT: F01Z5101BZ			 DPF CANNING: F01Z5017NP-00 DPF SLEEVE: F01Z5101BY DPF CATALYST: F01Z501890 DPF MAT: F01Z5101VW-00		
变更原因	降本更换催化剂衬垫;					
风险点 (影响)	更换衬垫催化剂有吹出风险;					
验证方法	Canning DV实验已通过;					

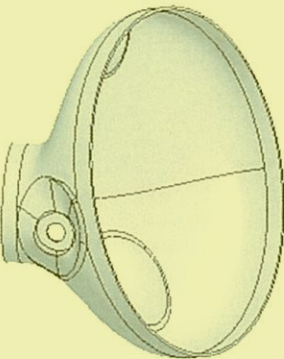
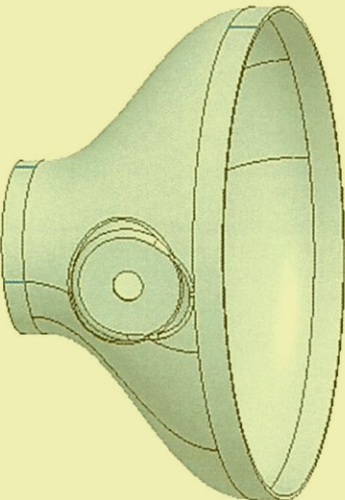
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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017XR-00	DPF BRACKET 1	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述	DPF BRACKET 1: F01Z5016MU DPF BRACKET: F01Z5101C0 WELDING NUT: F01Z500753 		DPF BRACKET 1: F01Z5017XR-00 DPF BRACKET: F01Z5102DJ-00 DPF BRACKET : F01Z5102DK-00 		
变更原因	根据客户边界重新设计DPF安装支架;				
风险点 (影响)	支架更改强度不满足设计要求;				
验证方法	B样放行前进行FEA仿真;				

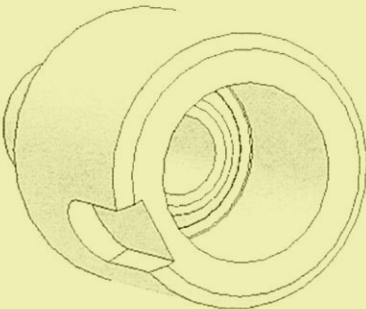
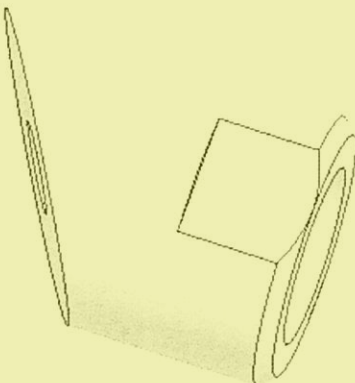
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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51025W-00	CONE	JIM-493	A	BOSCH/Customer
	变更前		变更后		

变更描述		
	重新设计DPF出气端锥；	
	变更原因	
	根据客户边界重新设计DPF出气端锥；	
风险点 (影响)	端锥形状及出口大小改变，气体流速均匀性降低；	
验证方法	进行CFD仿真，仿真报告如附件；	

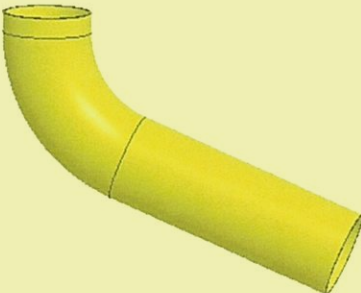
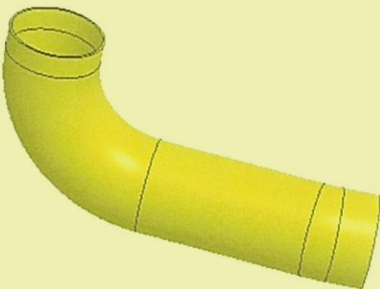
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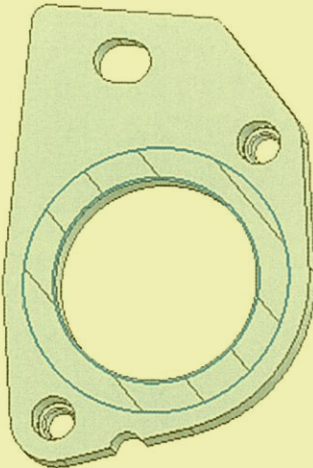
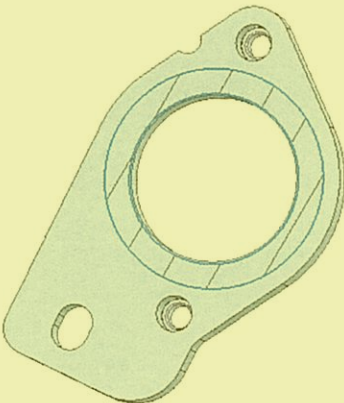
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51025X-00	DP SENSOR BOSS	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	根据客户边界重新设计压差管座;				
风险点 (影响)	无;				
验证方法	无;				

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	零件号	零件名称	Project	样件阶段	变更来源	
	F01Z5102D3-00	CONNECTION PIPE	JIM-493	A	BOSCH/Customer	
	变更前					
变更描述						
	根据客户边界重新设计DPF出气管；					
	根据客户边界重新设计DPF出气管；					
	管子走向改变，气体流速均匀性随之更改；					
变更原因	根据客户边界重新设计DPF出气管；					
风险点 (影响)	管子走向改变，气体流速均匀性随之更改；					
验证方法	进行CFD仿真，仿真报告如附件；					

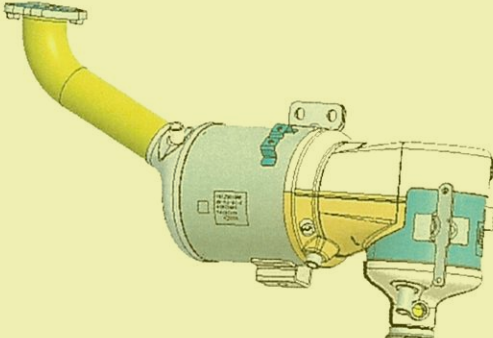
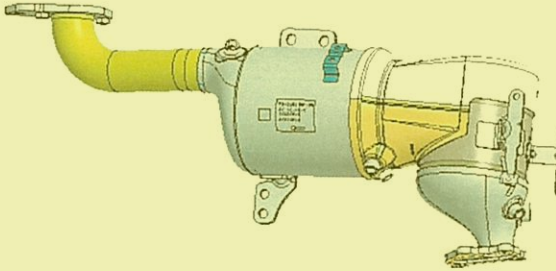
PDECR 24-093

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5102D3-00	CONNECTION PIPE	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	与4JJ法兰保持一致，借用4JJ法兰；				
风险点 (影响)	无；				
验证方法	无；				

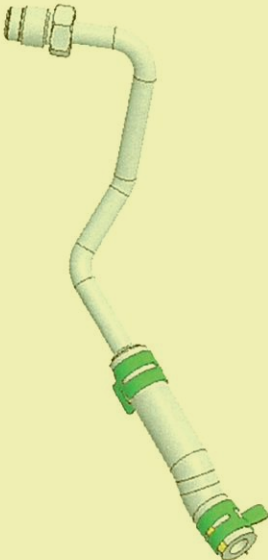
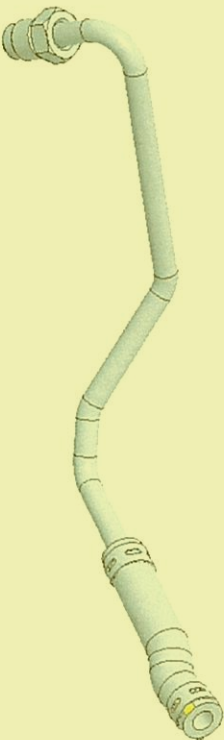
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PDECR 24-093

4

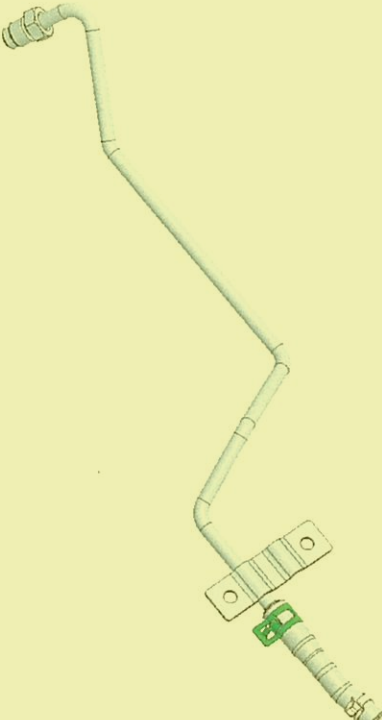
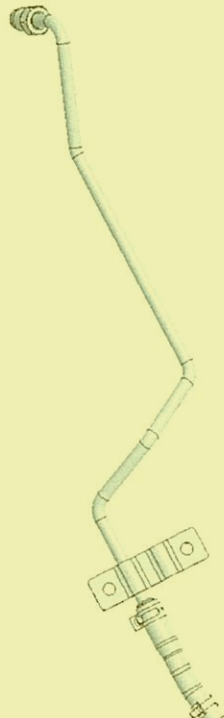
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017W5-00	DOC+SDPF WELDING	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	上述变更后，焊接组件随之更改；				
风险点（影响）	无				
验证方法	无				

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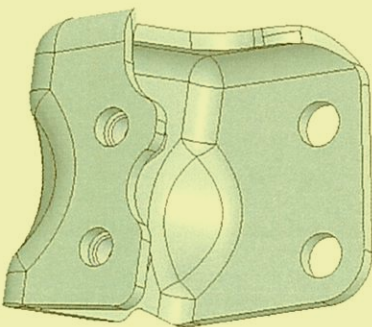
	零件号	零件名称	Project	样件阶段	变更来源
	See below	DP tube inlet	JIM-493	A	BOSCH/Customer
	变更前				
变更描述					
	DP tube inlet : F01 Z5016MZ DP tube inlet : F01 Z5101 C9		DP tube inlet : F01 Z501 7W8-00 DP tube inlet : F01 Z5102D5-00		
变更原因	根据客户边界更改压差管走向;				
风险点 (影响)	无				
验证方法	无				

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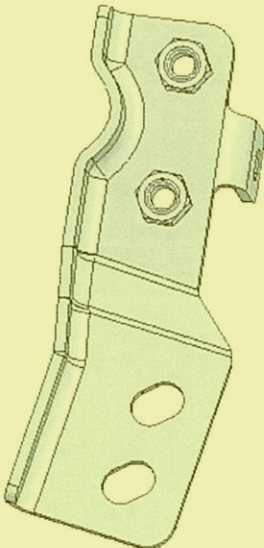
3

	零件号	零件名称	Project	样件阶段	变更来源
	See below	DP tube OUTlet	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	DP tube OUTlet : F01Z5016MY DP tube OUTlet : F01Z5101C8				
风险点 (影响)	DP tube OUTlet : F01Z5017W7-00 DP tube OUTlet : F01Z5102D4-00				
验证方法	根据客户边界更改压差管走向;				
	无				
	无				

PDECR 24-093

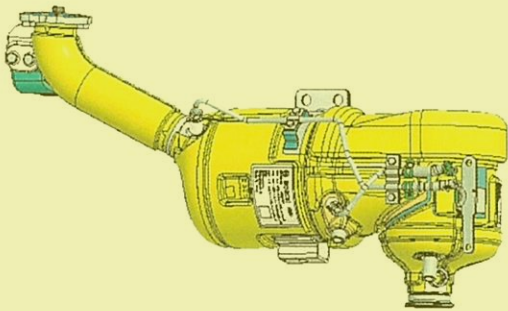
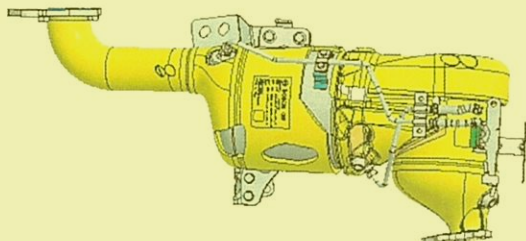
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017XS-00	DPF BRACKET	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述	无		 DPF BRACKET：F01Z5017XS-00 DPF BRACKET：F01Z51025B-00 NUT：F01Z560017		
变更原因	根据客户边界重新设计DPF转接支架；				
风险点（影响）	支架更改强度不满足设计要求；				
验证方法	B样放行前进行FEA仿真；				

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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017XT-00	DPF BRACKET	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述	无		 DPF BRACKET：F01Z5017XT-00 DPF BRACKET：F01Z5102D1-00 NUT：F01Z500753		
变更原因	根据客户边界重新设计DPF转接支架；				
风险点（影响）	支架更改强度不满足设计要求；				
验证方法	B样放行前进行FEA仿真；				

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3

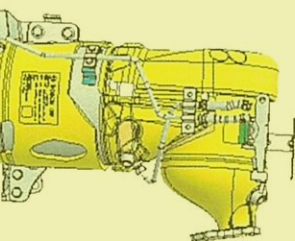
零件号	零件名称	Project	样件阶段	变更来源
See below	DOC+SDPF UNIT	JIM-493	A	BOSCH/Customer
变更前		变更后		
				
变更描述				
变更原因				
风险点 (影响)				
验证方法				

2

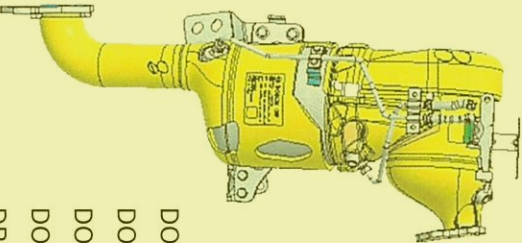
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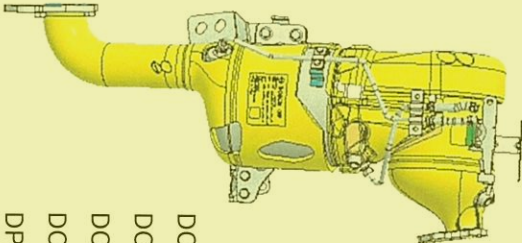
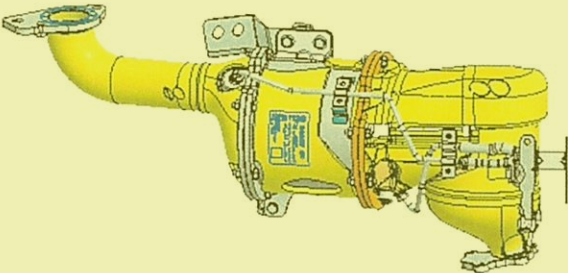
3

	零件号	零件名称	Project	样件阶段	变更来源
	See below	DOC+SDPF UNIT	JIM-493	A	BOSCH/Customer
	变更后				
变更描述	变更前				
	 DOC+SDPF UNIT: F01ZH003FU-01 DOC+SDPF WELDING: F01Z5017W5-00 DOC CANNING: F01Z5017NM-00 DOC CATALYST: F01Z50178N-00 DPF CANNING: F01Z5017NP-00 DPF CATALYST: F01Z501890				
变更原因	释放JIM 493 DOC+DPF老化件;				
风险点 (影响)	无				
验证方法	无				

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	零件号	零件名称	Project	样件阶段	变更来源
	See below	DOC+SDPF UNIT	JIM-493	A	BOSCH/Customer
	变更前				
变更描述	变更后				
变更原因					
	DOC+SDPF UNIT: F01ZH003FU-01 DOC+SDPF WELDING: F01Z5017W5-00 DOC CANNING: F01Z5017NM-00 DOC CATALYST: F01Z50178N-00 DPF CANNING: F01Z5017NP-00 DPF CATALYST: F01Z501890				
变更原因	DOC+SDPF UNIT: F01ZH003G1-00 DOC UNIT: F01Z5017X9-00 DOC WELDING: F01Z5017XA-00 DOC CATALYST: F01Z50178N-00 DPF UNIT : F01Z5017XC-00 DPF WELDING : F01Z5017XD-00 DOC CATALYST: F01Z501890 PIPE UNIT : F01Z5017XF-00 PIPE WELDING : F01Z5017XG-00				
风险点 (影响)	释放JIM 493 DOC+DPF可拆新鲜件 (增加法兰及密封垫) ;				
验证方法	无				

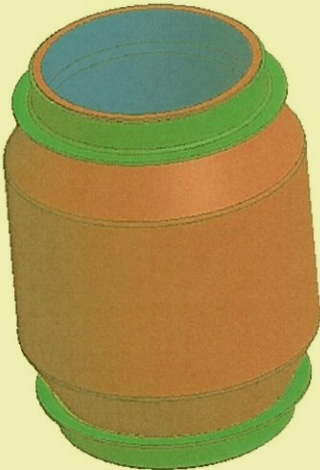
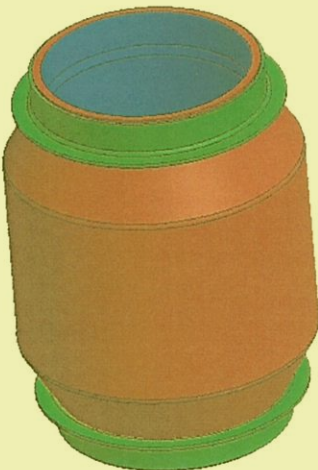
PDECR 24-093

	零件号	零件名称	Project	样件阶段	变更来源
	See below	DOC+SDPF UNIT	JIM-493	A	BOSCH/Customer
	变更前后				
变更描述	 DOC+SDPF UNIT: F01ZH003FU-51 DOC+SDPF WELDING: F01Z5017W5-51 DOC CANNING: F01Z5017NM-51 DOC CATALYST: F01Z50178T-00 DPF CANNING: F01Z5017NP-51 DPF CATALYST: 0452D318RJ DOC+SDPF UNIT: F01ZH003G1-51 DOC UNIT: F01Z5017X9-51 DOC WELDING: F01Z5017XA-51 DOC CATALYST: F01Z50178T-00 DPF UNIT : F01Z5017XC-51 DPF WELDING : F01Z5017XD-51 DOC CATALYST: 0452D318RJ PIPE UNIT : F01Z5017XF-00 PIPE WELDING: F01Z5017XG-00 				
变更原因	释放JIM 493 DOC+DPF可拆老化件 (增加法兰及密封垫) ;				
风险点 (影响)	无				
验证方法	无				

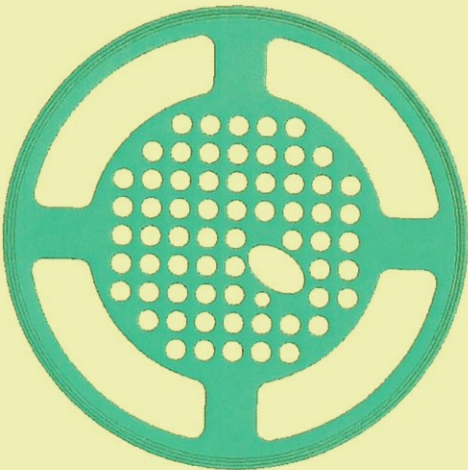
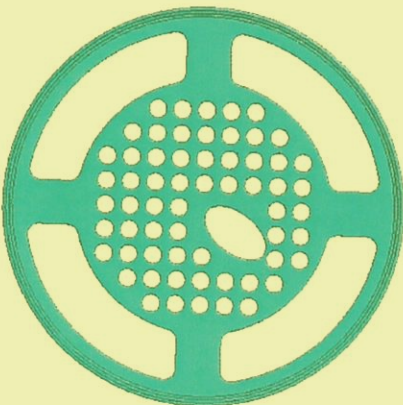
1

3

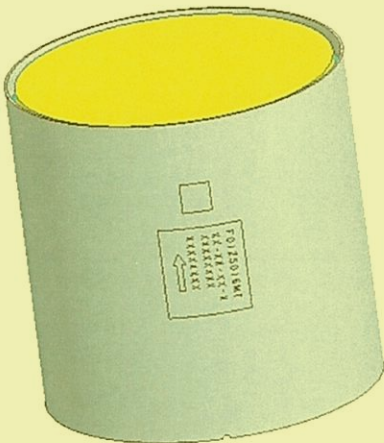
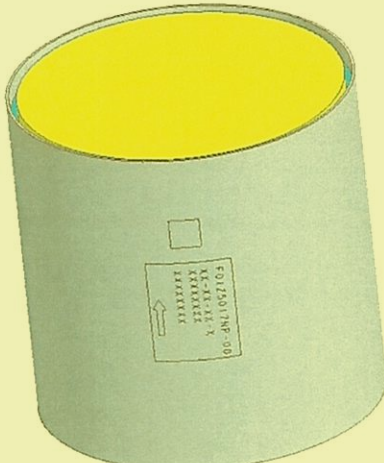
PDECR 24-093

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51023K-00	FLEX ELEMENT	JIM-493	A	BOSCH/Customer
变更描述	变更前		变更后		
					
	降本，波纹管换成腾龙的波纹管；				
变更原因	降本，波纹管换成腾龙的波纹管；				
风险点（影响）	波纹管强度不足；				
验证方法	后续做实验验证；				

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	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51020H-00	SCR INLAT PLATE	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述			 借用JP560现有扰流板;		
变更原因	借用JP560现有扰流板;				
风险点 (影响)	扰流板更改会影响气体流速均匀性;				
验证方法	进行CFD仿真, 仿真报告如附件;				

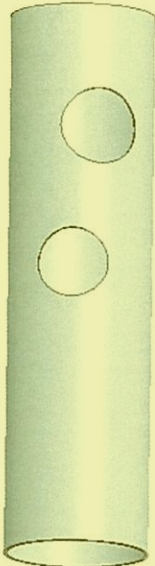
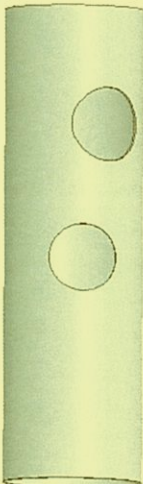
PDECR 24-093

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017NP-00	DPF CANNING	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述	 SCR CANNING: F01Z5016NA SCR SLEEVE: F01Z5101D2 SCR CATALYST: F01Z501891 SCR MAT: F01Z5101D3		 SCR CANNING: F01Z5017WA-00 SCR SLEEVE: F01Z5101D2 SCR CATALYST: F01Z501891 SCR MAT: F01Z5101PU-02		
变更原因	降本更换催化剂衬垫;				
风险点 (影响)	更换衬垫催化剂有吹出风险;				
验证方法	Canning DV实验已通过;				

1

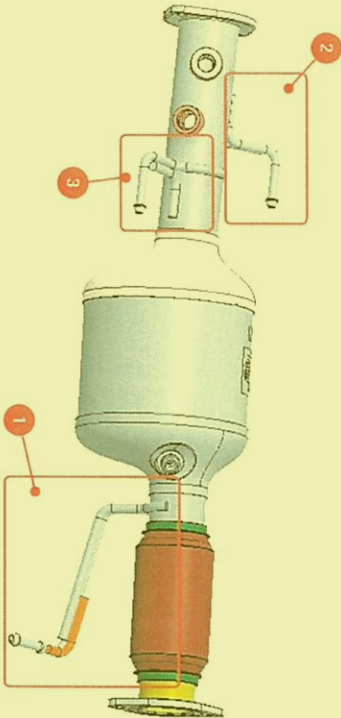
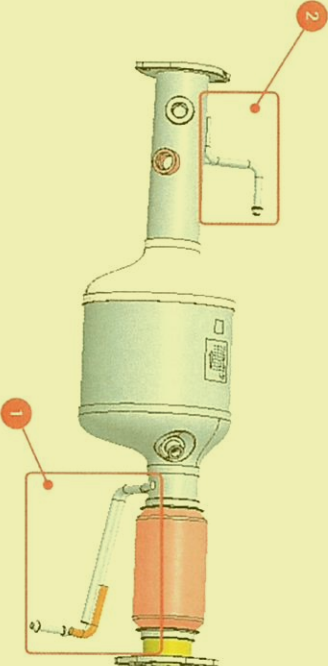
PDECR 24-093

2

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51025J-00	OUTLET PIPE	JIM-493	A	BOSCH/Customer
	变更前		变更后		
变更描述					
变更原因	根据客户边界缩短尾管长度;				
风险点 (影响)	无				
验证方法	无				

PDECR 24-093

1

		零件号	零件名称	Project	样件阶段	变更来源
		F01Z5017W9-00	SCR WELDING	JIM-493	A	BOSCH/Customer
		变更前		变更后		
变更描述						
变更原因	根据客户边界调整SCR数模;					
风险点 (影响)	取消一组吊钩, 强度不足;					
验证方法	B样放行前进行FEA仿真;					

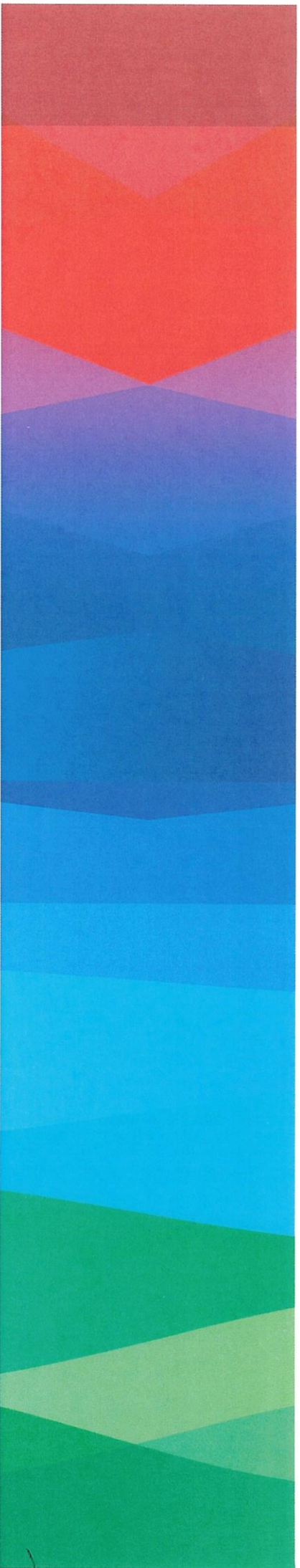
2

PDECR 24-093

	零件号		零件名称	Project	样件阶段	变更来源
	F01ZH003FV-01		SCR UNIT	JIM-493	A	BOSCH/Customer
			变更前	变更后		
变更描述						
变更原因	保温借用JP560弗吉亚开模保温;					
风险点 (影响)	无					
验证方法	无					

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	零件号		零件名称	Project	样件阶段	变更来源
	See below		SCR UNIT	JIM-493	A	BOSCH/Customer
	变更前					
变更描述	 SCR UNIT: F01ZH003FV-01 SCR WELDING: F01Z5017W9-00 SCR CANNING: F01Z5017WA-00 SCR CATALYST: F01Z501891			 SCR UNIT: F01ZH003FV-51 SCR WELDING: F01Z5017W9-51 SCR CANNING: F01Z5017WA-51 SCR CATALYST: 0452D5518B		
变更原因	释放JIM 493 SCR老化件;					
风险点 (影响)	无					
验证方法	无					



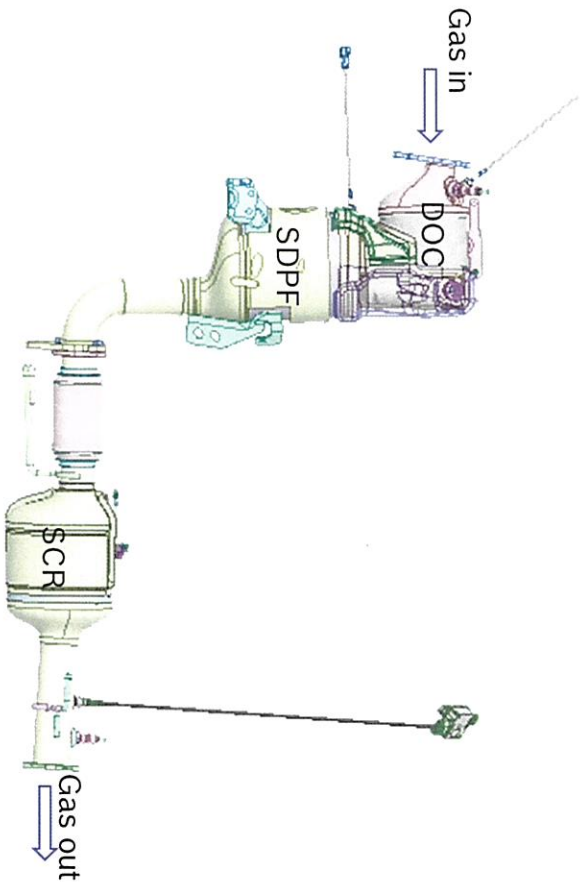
The JMC 493 CFD simulation report

For ATS-493-20240816.stp

The JMC493 CFD simulation report

Boundary Conditions

Simulation point	Mass flow [kg/h]	Temperature [°C]	Dosing (mg/s)	Outlet Pressure [Pa]
OP1	585	647	220	101325
OP2	268	266	60	101325



Catalyst information

DOC	Ø5.66"x4"	400/4
SDPF	Ø6.77"x7"	300/12
SCR	Ø6.5"x5"	400/4

2

Spray parameters

3

4 Spray Data ETI-3.6 (0280158708) @ 5 bar

5

→ Geometrical & functional data of nozzle holes

- Number: $n_H = 6$
- Hole diameter: $D_H \approx 130 \mu m$
- Diameter at hole centre positions: 1.9 mm
- Circumferential distribution: 60°
- Static mass flow: $\dot{m} = 5.5 \text{ kg/h}$
- Spray angle from visualization: $\alpha_{vis} = 22^\circ$

→ Proposal for CFD spray initialization¹⁾

- Equivalent spray type: 6 hole full cone spray
- Cone angle: $\beta = 7^\circ$
- Spray angle:²⁾ $\gamma = 9^\circ$
- Estimated initial droplet velocity: $v_{in} = 24 \text{ m/s}$
- Droplet Diameter, SMD: $D_{32} = 108 \mu m$

Remarks:

1) Only for simulation purpose, deviations in series production not considered!

2) Total spray angle ($2\gamma+\beta$) can differ to visual spray angle because of geometrical projection & different measurement techniques.

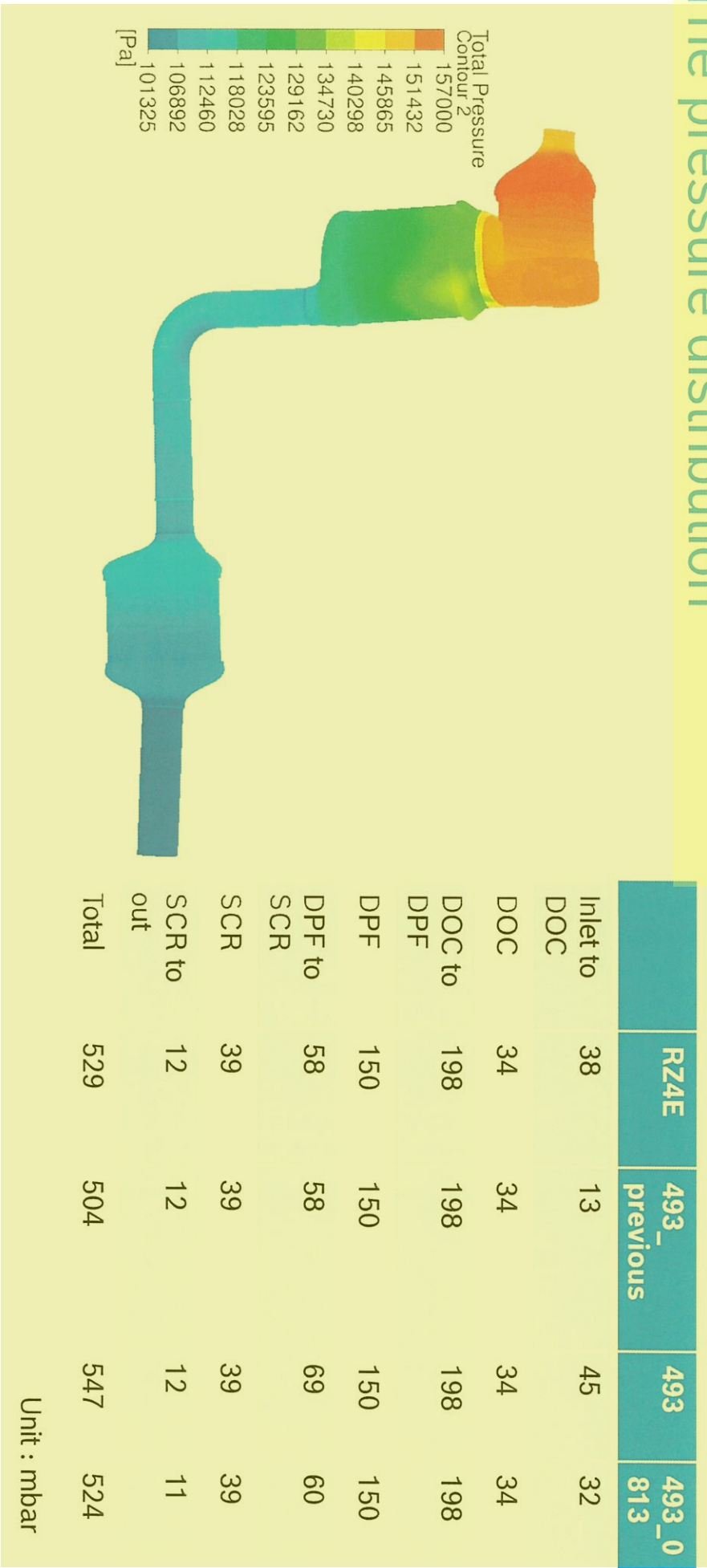
The total pressure drop of the ATS is 529mbar at the OP1 (fresh SDPF)

3

The JMC493 CFD simulation report

The pressure distribution

2

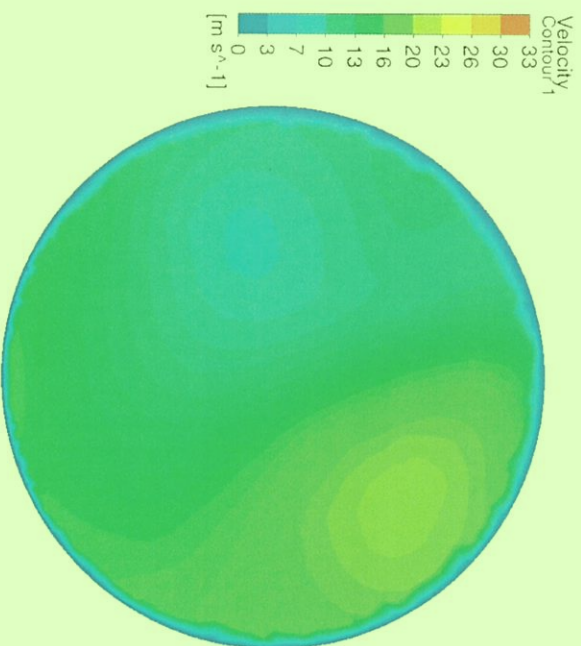


The JMC493 CFD simulation report

The velocity uniformity@OP1

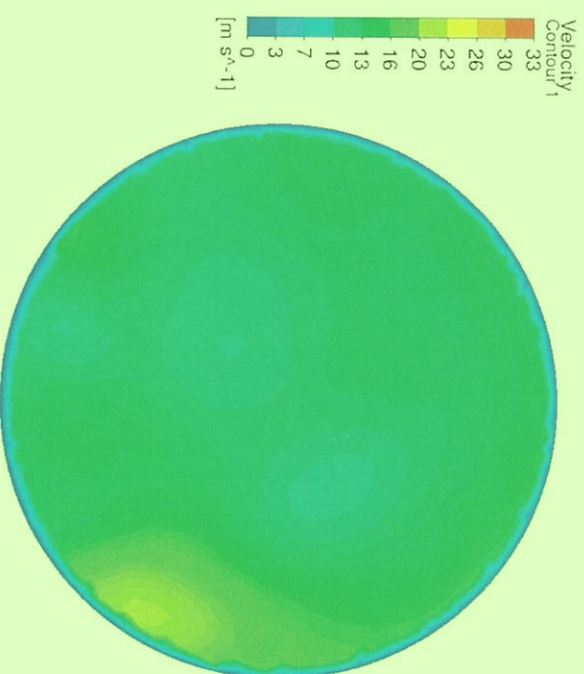


DOC inlet



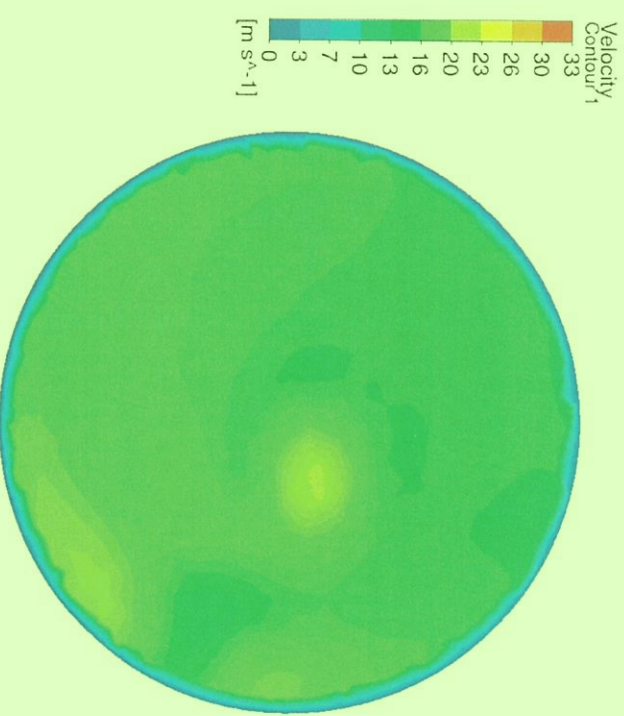
$U_{vel} = 0.95$

SDPF inlet



$U_{vel} = 0.97$

SCR inlet



$U_{vel} = 0.96$

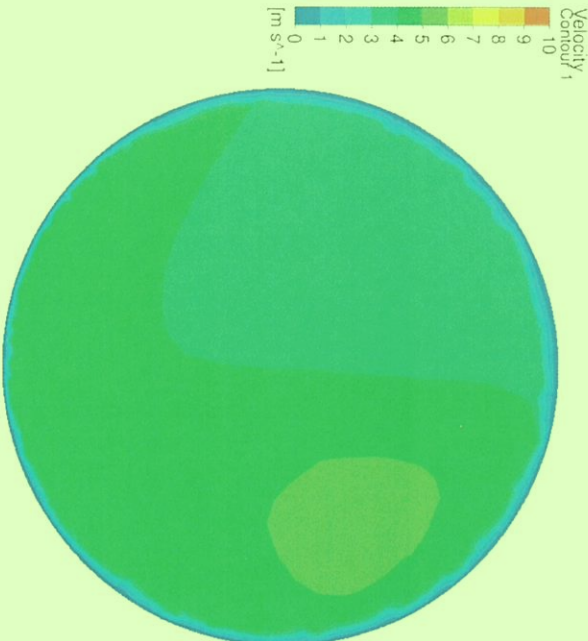
Velocity uniformity index at the DOC inlet, SDPF inlet, SCR inlet all can meet the design target ($U_{I} \geq 0.95$)

The JMC493 CFD simulation report

The velocity uniformity@OP2

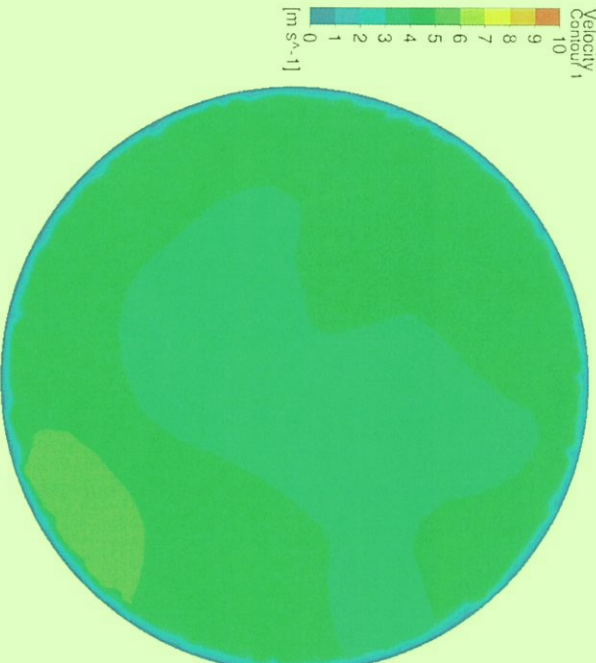


DOC inlet



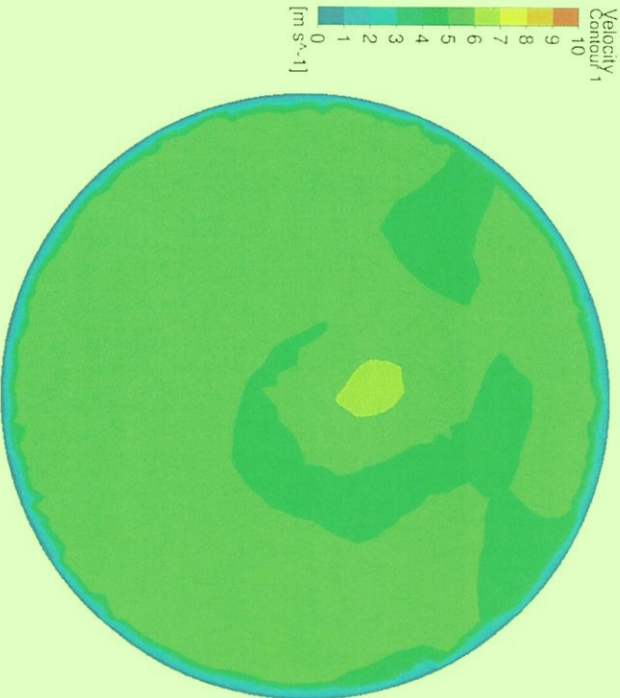
$U_{vel} = 0.96$

SDPF inlet



$U_{vel} = 0.98$

SCR inlet



$U_{vel} = 0.97$

Velocity uniformity index at the DOC inlet, SDPF inlet , SCR inlet all can meet the design target ($U_{I} \geq 0.95$)



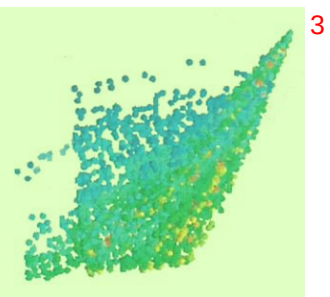
1

The JMC493 CFD simulation report

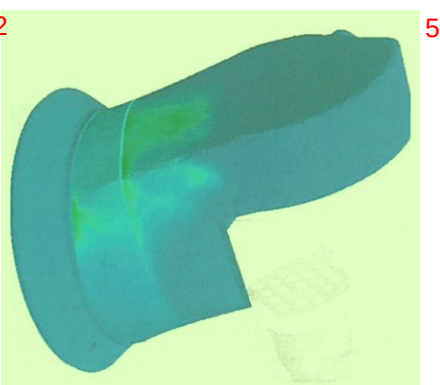
The spray visualization & impingement



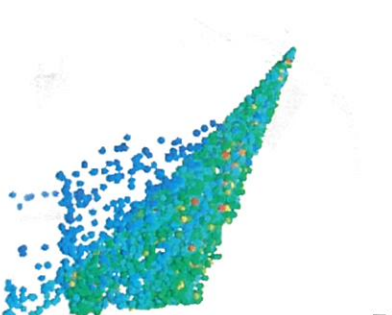
OP1



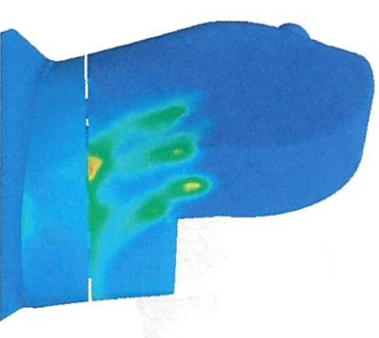
3



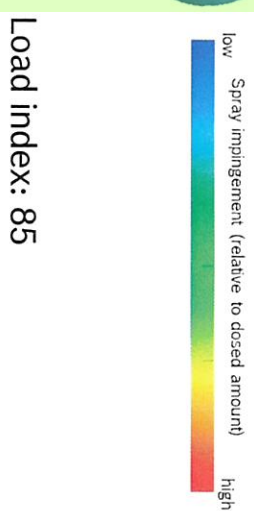
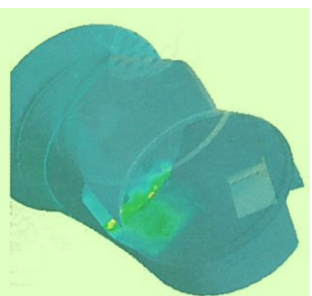
5



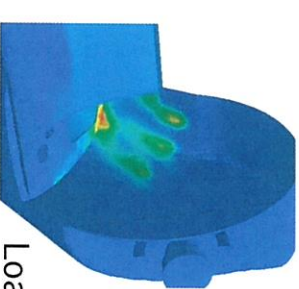
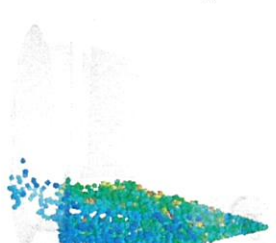
OP2



2



Load index: 85



Load index: 100

The spray droplet will not knock on DOC or flow through mix plate to SDPF directly, the spray load index value lower than the experience value(130), which means the crystallization risk is not too high.

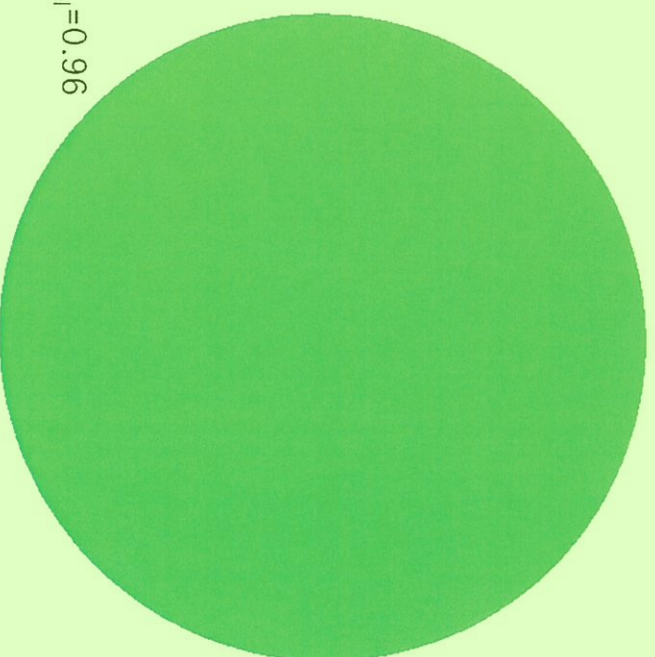
The JMC493 CFD simulation report

Ammonia equivalent distribution-SDPF inlet



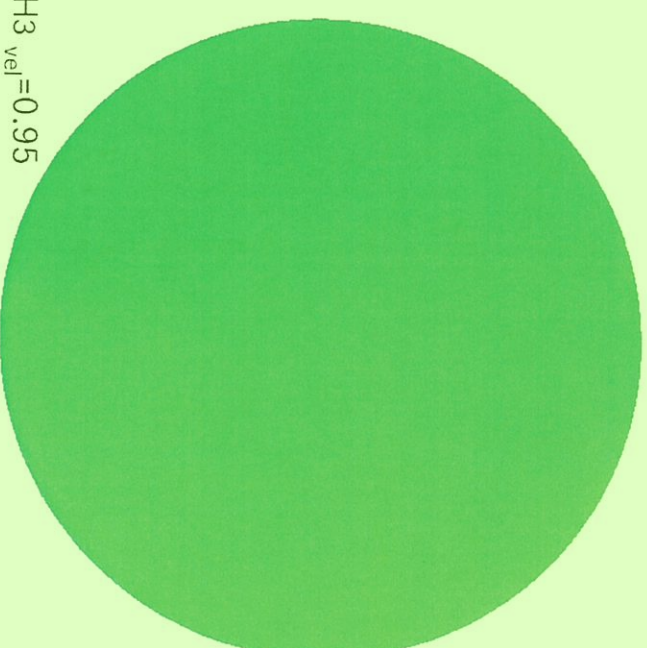
View direction
→

OP1



$\text{NH}_3_{\text{vel}} = 0.96$

OP2



$\text{NH}_3_{\text{vel}} = 0.95$



The ammonia equivalent mass fraction uniformity at the SDPF above the target index value 0.95 at both OP

7

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The JMC493 CFD simulation report

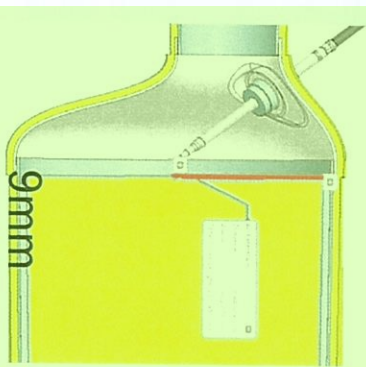
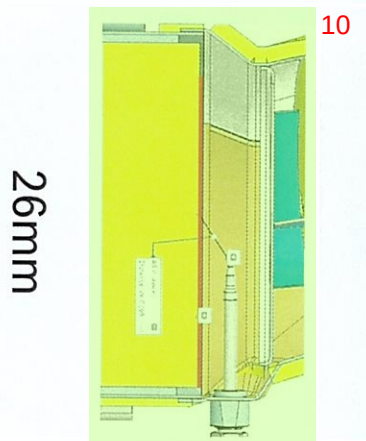
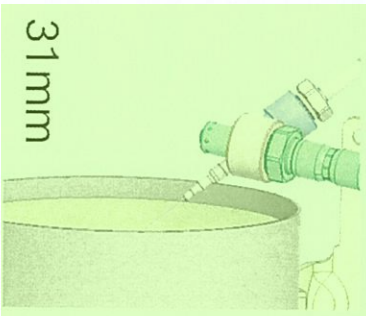
Temperature sensor position

T4

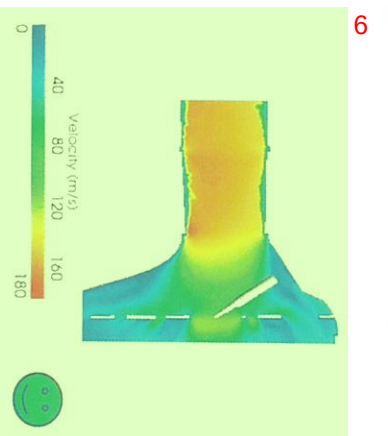
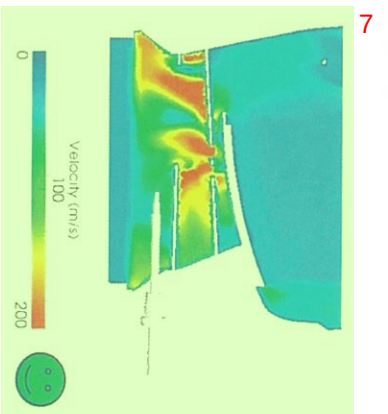
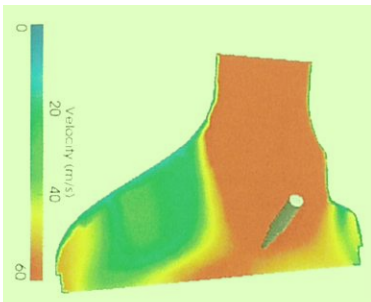
T5

T6

Distance from sensor tip to catalyst center (<80mm)



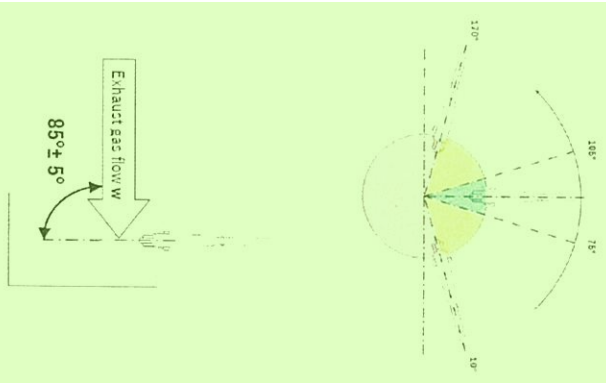
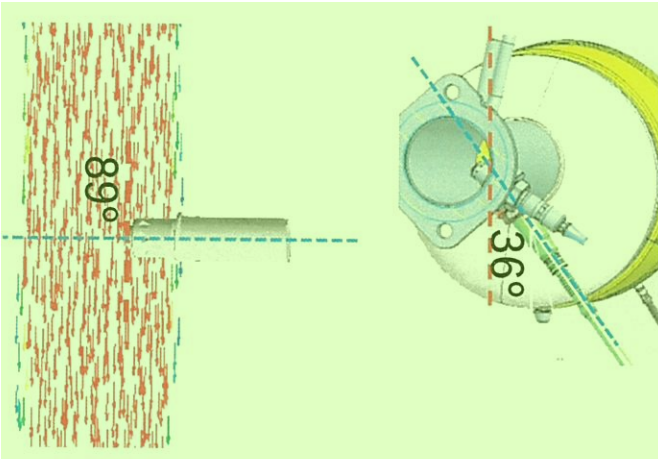
The velocity distribution around the sensor tip (At the mainstream area)



The JMC493 CFD simulation report

Sensor mounting position accuracy-PM sensor



BOSCH SPEC	Simulation result	Pass or not
1  2	4 	Pass Pass

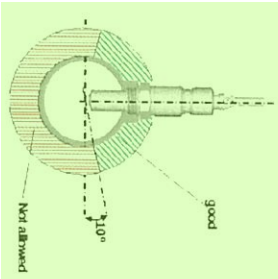
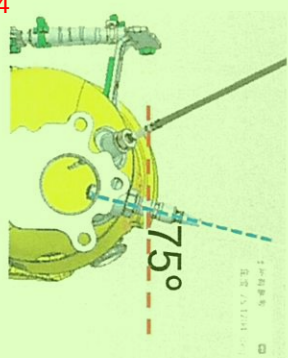
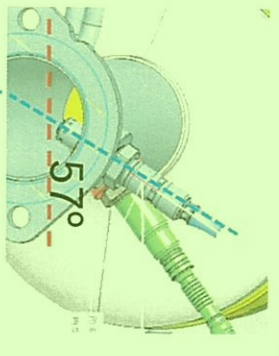
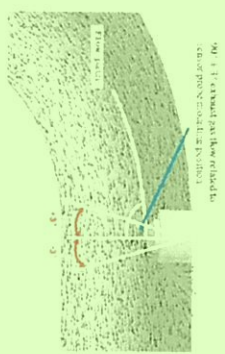
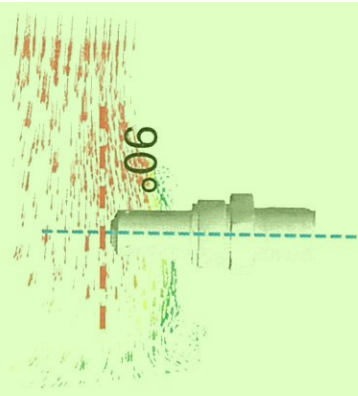
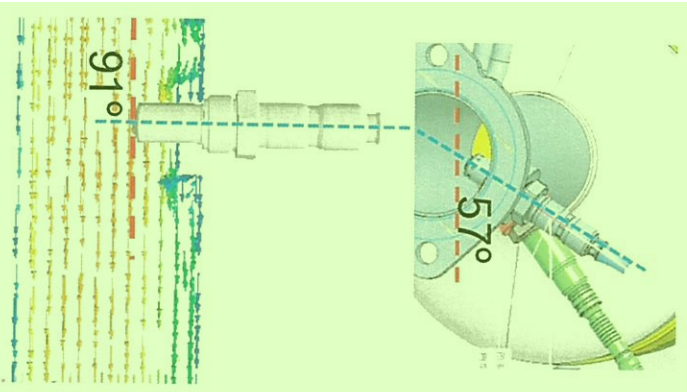
Background

- One PM sensor planned mounted on the backpipe

Objectives

- Using CFD simulation to analyze the position for the PM sensor by using the criteria according to the "Exhaust gas sensor layout requirement"

The JMC493 CFD simulation report Sensor mounting position accuracy-Nox sensor

BOSCH SPEC		Simulation result		Pass or not
		Upstream	Downstream	
1				Pass
2				Pass

Background

- Two NO_x sensors are planned (one upstream the DOC and one downstream the SCR catalyst).

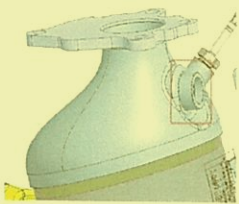
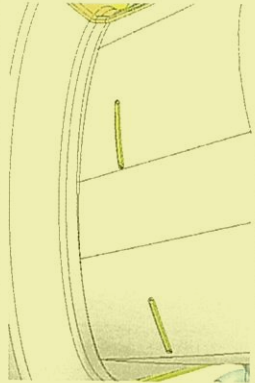
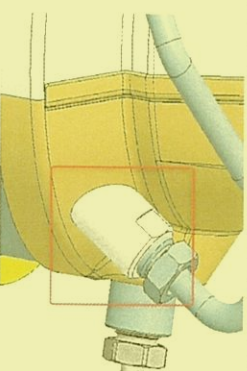
Objectives

- Using CFD simulation to analyze the position for the NO_x sensor by using the criteria according to the “Exhaust gas sensor layout requirement”

The JMC493 CFD simulation report

Summary

- The back pressure of the whole system is 529mbar with fresh SDPF.
- Velocity uniformity at DOC inlet, SDPF inlet, SCR inlet all can meet the design target (≥ 0.95).
- Ammonia uniformity at SDPF inlet can meet the internal target value (≥ 0.95).

Basic information 基本信息		JIM 493		Meeting information 会议信息		DFMA Participants 参会人	
Project name 项目名称	Sample phase 样件阶段	Plant/Supplier 工厂/供应商	Meeting time 会议时间	Chen Maolian; Li Yang; Zhu Wenwu; Lu Qingsong; Fang Jiayan; Yin Kangzhuang; Yang Guangtuo; Zhang Bo			
JIM 493	A-sample	弗吉亚	2024.08.15 14:00-16:00				
Check point 检查项							
Welding feasibility 焊接可行性	Assembly feasibility 装配可行性	Measure feasibility 测量可行性	Parts feasibility 零件生产可行性				
Open Topic 会议讨论点							
Description 文字描述	Picture 图片		Action 下一步计划				
DOC进气端锥NOX座焊接干涉;			1.调整位置, 看下面是否修改数模; 2.弗吉亚评估能否通过调整工艺尽量不修改模型;				
混合器隔板不好焊接			评估能否把混合器隔板从内插改为翻边;				
混合器压差管角度大小无法焊接;			弗吉亚评估内焊可行性;				
	Jin Wei	6月30	装配可行性	Ongoing			